

FUNCTIONAL DESIGN SPECIFICATION

INDUSTRIAL BOTTLE PACKAGING LINE AUTOMATION

Document Information

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System Information

Item	Description
PLC	Allen-Bradley MicroLogix 1100
Programming Software	RSLogix 500
PLC Emulator	RSLogix Emulate 500
Communication Driver	RSLink Classic
HMI Software	Wonderware InduSoft Web Studio (AVEVA Edge)
Programming Language	Ladder Logic (LAD)
Application	Industrial Bottle Packaging Line Automation

Document Purpose

This Functional Design Specification (FDS) defines the functional requirements, operating philosophy, system architecture, and design objectives of the **Industrial Bottle Packaging Line Automation** project.

The purpose of this document is to describe **what the system is designed to accomplish** from a functional perspective. It serves as the primary engineering reference for understanding the intended behaviour of the automated bottle packaging line before reviewing the PLC program or HMI implementation.

This document does not describe the internal PLC programming logic in detail. Software implementation details, ladder logic architecture, and routine-level descriptions are documented separately within the **Software Design Specification (SDS)**.

Document Scope

This document covers the functional design of the following system operations:

- Automatic bottle transportation
 - Bottle filling process
 - Leak bottle detection
 - Bottle capping
 - Cap presence verification
 - Combined sorting and reject operation
 - Alarm management
 - Production monitoring
 - Runtime calculation
 - Manual fault simulation
 - Emergency Stop functionality
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Intended Audience

This document is intended for:

- Controls Engineers
 - PLC Programmers
 - Automation Engineers
 - System Integrators
 - Students learning Industrial Automation
 - Technical Reviewers
 - Recruiters evaluating automation projects
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Document Classification

Project Type: Proof of Concept (PoC)

This project has been developed as an educational and portfolio demonstration of industrial automation principles. While the machine closely represents the workflow of a real bottle packaging line, it is not intended to be a complete production-ready industrial solution. Additional safety systems, industrial communication networks, advanced diagnostics, and production management features can be incorporated for commercial applications.

Prepared By

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1. Introduction

1.1 Purpose of the Document

The purpose of this Functional Design Specification (FDS) is to define the functional requirements, operating philosophy, and overall system design of the **Industrial Bottle Packaging Line Automation** project.

This document serves as the primary engineering reference for understanding the intended operation of the automated system before implementation. It describes the functions that the machine is expected to perform, the interaction between different process stations, and the operational requirements necessary for safe and reliable system performance.

The document focuses on the functional behaviour of the machine rather than the detailed software implementation. PLC program architecture, ladder logic routines, memory allocation, and HMI implementation are documented separately within their respective engineering documents.

1.2 Project Background

Industrial bottle packaging systems are widely used in the food, beverage, pharmaceutical, and chemical industries to automate repetitive manufacturing processes while maintaining consistent product quality and production efficiency.

A typical packaging line performs multiple sequential operations, including bottle transportation, liquid filling, cap application, product inspection, sorting, and rejection of defective products. These operations are coordinated by a Programmable Logic Controller (PLC), which continuously monitors field devices and executes the required control logic.

This project has been developed to demonstrate the implementation of these industrial automation principles using an Allen-Bradley MicroLogix 1100 PLC programmed in RSLogix 500, with operator interaction provided through Wonderware InduSoft Web Studio (AVEVA Edge). The machine operation follows the sequence of bottle transportation, filling, capping, quality verification, sorting, and production monitoring, while incorporating fault handling and alarm management to simulate realistic industrial operating conditions.

1.3 Project Scope

The scope of this project includes the design, implementation, simulation, and visualization of an automated bottle packaging process.

The system has been developed to perform the following primary functions:

- Automatic bottle transportation between processing stations.

- Automatic bottle filling to a configurable preset level.
- Leak bottle identification following the filling process.
- Automatic bottle capping.
- Cap presence verification.
- Combined sorting and reject operation.
- Production counting and runtime monitoring.
- Alarm generation and management.
- Manual fault simulation through the HMI.
- Emergency Stop functionality for safe machine shutdown.

The project demonstrates the complete control philosophy of an automated bottle packaging line within a simulated environment.

1.4 Project Objectives

The primary objectives of this project are to:

- Develop a modular PLC program using industrial programming practices.
 - Simulate an automated bottle packaging process using Ladder Logic.
 - Design an operator-friendly Human-Machine Interface (HMI).
 - Demonstrate sequential machine control using state-based operation.
 - Implement automatic filling and capping processes.
 - Detect defective bottles through leak and cap verification.
 - Reject defective products while allowing acceptable products to continue.
 - Monitor production performance through real-time statistics.
 - Provide alarm handling and fault recovery mechanisms.
 - Demonstrate industrial automation concepts suitable for educational and portfolio purposes.
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1.5 System Description

The Industrial Bottle Packaging Line consists of multiple process stations that operate together to complete a single production cycle for each bottle.

The production process begins with bottle transportation using the infeed conveyor. Bottles are automatically positioned at the filling station, where liquid is dispensed until the configured filling level is achieved. Following the filling operation, the bottle proceeds to the capping station, where a cap is applied before moving to the combined sorting and reject station.

At the final station, the PLC evaluates the quality of the processed bottle by verifying cap presence and checking whether the bottle has been identified as leaking. Acceptable bottles continue toward the outfeed conveyor, while defective bottles are removed from the production line using the reject mechanism.

Throughout the production process, the PLC continuously updates production statistics, machine runtime, alarm status, and system operating conditions, all of which are displayed through the HMI for operator monitoring.

1.6 Proof of Concept Statement

This project has been developed as a **Proof of Concept (PoC)** to demonstrate industrial automation principles, PLC programming techniques, and HMI development within a simulated production environment.

Although the project closely represents the operation of a real industrial bottle packaging line, it is not intended to replicate every feature found in commercial manufacturing systems. Real production facilities typically include additional safety hardware, industrial communication networks, servo motion control, advanced diagnostics, production traceability, and manufacturing execution systems that are beyond the scope of this implementation.

The primary objective of this project is to demonstrate the design methodology, modular software architecture, and control philosophy commonly applied in industrial automation projects.

2. Functional Requirements

2.1 General Requirements

The Bottle Packaging Line shall operate as a fully automated system capable of transporting, filling, capping, inspecting, sorting, and monitoring bottles while providing operators with real-time machine status through the Human-Machine Interface (HMI).

The PLC shall continuously monitor all process inputs, execute the required control logic, and operate field devices in accordance with the defined control philosophy.

The machine shall be capable of operating continuously under normal conditions while providing safe responses to abnormal operating situations.

2.2 Bottle Transportation Requirements

FR-001 – Bottle Transportation

The system shall automatically transport bottles between all process stations using the conveyor system.

FR-002 – Bottle Positioning

The system shall accurately position bottles at the filling, capping, and sorting stations before initiating any processing operation.

FR-003 – Controlled Movement

Bottle movement shall be controlled by the PLC to ensure that only one bottle is processed at each station during a production cycle.

2.3 Filling Requirements

FR-004 – Filling Operation

The system shall automatically fill each bottle once it has been correctly positioned at the filling station.

FR-005 – Preset Fill Level

The filling process shall continue until the configured preset filling level has been achieved.

FR-006 – Filling Control

The filling valve shall remain energized only while filling is required and shall close immediately after the target level is reached.

FR-007 – Filling Parameters

The operator shall be able to configure the desired filling level through the HMI.

2.4 Leak Detection Requirements

FR-008 – Leak Bottle Identification

The system shall provide the capability to identify bottles that are considered defective due to leakage.

FR-009 – Product Tracking

Once a bottle has been identified as defective, the PLC shall retain this information until the bottle reaches the Combined Sorting and Reject Station.

FR-010 – Leak Bottle Handling

Leak bottles shall not be immediately removed from the production line. Instead, they shall continue through the process before being rejected at the Combined Sorting and Reject Station.

2.5 Capping Requirements

FR-011 – Cap Application

The system shall automatically apply a cap after the bottle has completed the filling operation.

FR-012 – Cap Press Operation

The cap press shall complete one full operating cycle consisting of extension and retraction before the bottle proceeds to the next station.

FR-013 – Home Position Verification

The PLC shall verify that the cap press has returned to its home position before allowing the production cycle to continue.

2.6 Quality Verification Requirements

FR-014 – Cap Presence Verification

The system shall verify that a cap has been successfully applied before the bottle exits the production process.

FR-015 – Product Quality Evaluation

The system shall determine whether each bottle satisfies the required quality conditions before it is accepted.

FR-016 – Defective Product Identification

Bottles identified with missing caps or leak conditions shall be classified as defective products.

2.7 Sorting and Reject Requirements

FR-017 – Bottle Sorting

The Combined Sorting and Reject Station shall evaluate every processed bottle before it leaves the production line.

FR-018 – Acceptable Products

Acceptable bottles shall continue toward the outfeed section.

FR-019 – Reject Operation

Defective bottles shall be removed from the production line using the reject mechanism.

FR-020 – Reject Verification

The reject mechanism shall complete its operating cycle before normal production resumes.

2.8 Production Monitoring Requirements

FR-021 – Production Counter

The system shall maintain a count of all successfully processed bottles.

FR-022 – Reject Counter

The system shall maintain a count of all rejected bottles.

FR-023 – Runtime Monitoring

The system shall continuously calculate and display machine runtime.

FR-024 – Bottles Per Hour

The system shall calculate production rate in bottles per hour and update the value every minute.

FR-025 – Production Statistics

Production statistics shall be available for operator viewing through the HMI.

2.9 Alarm Requirements

FR-026 – Alarm Detection

The PLC shall continuously monitor the machine for abnormal operating conditions.

FR-027 – Alarm Display

All active alarms shall be displayed on the HMI.

FR-028 – Alarm Acknowledgement

The operator shall be able to acknowledge alarms through the HMI.

FR-029 – Alarm Reset

The system shall require the operator to reset the alarm before returning to automatic operation.

2.10 Manual Simulation Requirements

FR-030 – Fault Simulation

The HMI shall allow manual simulation of machine faults for testing and demonstration purposes.

The following simulated faults shall be available:

- Leak Bottle Fault
- Cap Missing Fault
- Capping Arm Retract Jam
- Capping Arm Complete Jam
- Sorting Arm Retract Jam
- Sorting Arm Complete Jam

FR-031 – Simulation Recovery

The operator shall be able to clear simulated faults through the HMI after the required reset procedure has been completed.

2.11 Emergency Stop Requirements

FR-032 – Emergency Stop

Activation of the Emergency Stop shall immediately stop all machine operations.

FR-033 – Output De-energization

All outputs shall be de-energized while the Emergency Stop condition is active.

FR-034 – Controlled Restart

The machine shall not restart automatically after the Emergency Stop has been released. The operator shall perform the required alarm reset and issue a new Start command before production resumes.

2.12 Human-Machine Interface Requirements

FR-035 – Process Visualization

The HMI shall provide a graphical representation of the complete bottle packaging process.

FR-036 – Production Monitoring

The HMI shall display real-time production information, machine status, and performance indicators.

FR-037 – Machine Configuration

Authorized operators shall be able to modify production settings through the Configuration Screen.

FR-038 – Alarm Management

The HMI shall provide alarm indication, acknowledgement, silence, and reset functions.

FR-039 – User Authentication

Configuration functions shall be protected through user authentication to prevent unauthorized modification of machine settings.

3. Operating Philosophy

3.1 General Philosophy

The Industrial Bottle Packaging Line has been designed to operate as a fully automated sequential machine. The PLC continuously monitors process conditions and controls each stage of the production cycle to ensure that bottles are processed in the correct order while maintaining safe and reliable operation.

The machine follows a state-based operating philosophy in which only one major processing operation is performed at a time for each bottle. Each station completes its assigned task before the bottle is transferred to the next station. This approach simplifies process control, improves reliability, and reduces the possibility of incorrect machine operation.

During normal production, the PLC continuously evaluates sensor feedback, operator commands, and machine status to determine when the system should advance to the next stage of the process.

3.2 Automatic Operation

Automatic Mode is the primary operating mode of the machine.

After the operator initiates production using the Start command, the PLC automatically controls the complete bottle packaging process without requiring further operator intervention.

During Automatic Operation, the system performs the following functions:

- Transports bottles between all process stations.
- Controls bottle positioning.
- Executes the filling operation.
- Performs bottle capping.
- Verifies product quality.
- Rejects defective bottles.
- Updates production statistics.
- Monitors alarms and machine status.

The production cycle continues until the operator stops the machine or a fault condition interrupts normal operation.

3.3 Operator Interaction

The operator interacts with the machine through the Human-Machine Interface (HMI).

The HMI provides access to:

- Machine Start and Stop controls.
- Production status.
- Runtime information.
- Bottle counters.
- Reject counters.
- Bottles-per-hour calculation.
- Alarm management.
- Production configuration.
- Manual fault simulation.
- User authentication.

The operator is responsible for monitoring machine performance and responding appropriately to alarms when abnormal operating conditions occur.

3.4 Sequential Operation

The Bottle Packaging Line is designed as a sequential process where each bottle progresses through a series of predefined processing stages.

Each stage must complete successfully before the next stage is initiated. This ensures consistent product quality and prevents process conflicts between different stations.

The PLC verifies completion of each operation before permitting the bottle to continue through the production line.

3.5 Product Quality Philosophy

Every bottle processed by the system is evaluated before leaving the production line.

Product quality is determined using the following inspection criteria:

- Bottle successfully filled.
- No leak bottle condition detected.

- Cap successfully applied.
- Cap presence confirmed.

Any bottle that does not satisfy these quality requirements is classified as a defective product and is rejected at the Combined Sorting and Reject Station.

This philosophy ensures that only acceptable products continue toward the outfeed section.

3.6 Fault Philosophy

The automation system continuously monitors machine conditions throughout production.

Whenever an abnormal condition is detected, the PLC performs the appropriate corrective action based on the type of fault.

Fault handling may include:

- Rejecting the affected bottle.
- Stopping the affected device.
- Stopping the complete production line.
- Displaying an alarm message.
- Activating the alarm indicator.
- Requiring operator acknowledgement.

The objective is to protect the machine, maintain operator awareness, and prevent defective products from being accepted.

3.7 Manual Fault Simulation Philosophy

To support testing and educational demonstrations, the HMI provides manual fault simulation functions.

These functions allow operators to intentionally generate predefined fault conditions without requiring actual hardware failures.

The simulated faults enable verification of:

- PLC control logic.
- Alarm generation.
- HMI indication.
- Fault recovery procedures.

- Reject logic.
- Machine response.

Manual fault simulation is intended for commissioning, testing, and educational purposes only.

3.8 Production Monitoring Philosophy

The PLC continuously collects production data throughout machine operation.

The following production information is available through the HMI:

- Total bottles processed.
- Total rejected bottles.
- Machine runtime.
- Bottles per hour.
- Current operating status.
- Active alarms.

Production performance indicators are updated automatically during machine operation to provide operators with real-time information regarding machine productivity.

3.9 System Recovery Philosophy

Following removal of a fault condition, the operator shall verify that the machine is safe to restart.

Before production resumes:

- The fault condition shall be cleared.
- The alarm shall be acknowledged and reset.
- The Emergency Stop, if activated, shall be released.
- A new Start command shall be issued by the operator. (In case of The Emergency Stop)

In standard industrial practice, machines are not allowed to restart automatically after fault recovery, as this ensures operator safety and prevents unintended operation. However, in this educational setup, to facilitate faster testing and fault analysis, the system is designed such that an alarm reset resumes the process from the point at which it was paused.

3.10 Proof of Concept Philosophy

This project has been developed as a Proof of Concept (PoC) to demonstrate industrial automation concepts using PLC and HMI technologies.

The implemented control philosophy reflects common practices used in industrial bottle packaging systems while remaining suitable for simulation and educational use.

Although the project demonstrates realistic machine operation, it does not include every feature typically found in commercial production equipment. Additional functionality such as safety PLCs, industrial networking, servo motion control, vision inspection, production databases, and manufacturing execution systems can be incorporated in future developments.

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4. System Description

4.1 System Overview

The Industrial Bottle Packaging Line is an automated production system designed to simulate the complete packaging process of liquid-filled bottles. The system integrates bottle transportation, filling, capping, product inspection, sorting, reject handling, production monitoring, and alarm management into a single automated process.

The system is controlled by an Allen-Bradley MicroLogix 1100 PLC, which continuously monitors process inputs, executes control logic, and operates field devices according to the defined control philosophy. Operator interaction is provided through a Wonderware InduSoft Web Studio (AVEVA Edge) Human-Machine Interface (HMI), allowing real-time monitoring, configuration, and fault management.

The machine has been designed using a modular control philosophy, allowing each functional section to operate as part of an integrated production process while maintaining clear separation of machine responsibilities.

4.2 Process Stations

The Bottle Packaging Line consists of four primary processing stations, each performing a specific operation within the production cycle.

Bottle Transportation

The conveyor system transports bottles between all processing stations.

It is responsible for:

- Transporting bottles through the production line.
- Positioning bottles at each station.
- Maintaining continuous product flow.
- Coordinating bottle movement with PLC commands.

The conveyor only operates when the PLC determines that it is safe for the bottle to move to the next process stage.

Filling Station

The Filling Station performs the automatic liquid filling operation.

Its primary functions include:

- Detecting bottle presence.
- Positioning the bottle correctly.
- Dispensing liquid into the bottle.
- Monitoring the filling level.
- Completing filling at the configured preset value.
- Identifying leak bottle conditions when simulated.

The filling operation represents the first processing stage of the production cycle.

Capping Station

The Capping Station applies a cap to each filled bottle.

Its responsibilities include:

- Bottle positioning.
- Cap application.
- Capping actuator movement.
- Home position verification.
- Detection of capping arm faults.

The PLC verifies completion of the capping cycle before permitting the bottle to continue toward the next station.

Combined Sorting and Reject Station

The Combined Sorting and Reject Station performs the final product quality evaluation.

This station is responsible for:

- Verifying cap presence.
- Evaluating leak bottle status.
- Determining product acceptance.
- Rejecting defective bottles.
- Allowing acceptable bottles to continue through the production line.

Product quality evaluation is completed before the bottle exits the system.

4.3 Control System

The control system consists of a Programmable Logic Controller (PLC) and a Human-Machine Interface (HMI).

The PLC is responsible for:

- Reading field inputs.
- Processing control logic.
- Controlling machine outputs.
- Managing production sequencing.
- Detecting fault conditions.
- Updating production statistics.

The HMI provides:

- Machine status visualization.
- Operator controls.
- Alarm management.
- Production monitoring.
- Configuration settings.
- Manual fault simulation.

Together, the PLC and HMI provide complete supervision and control of the bottle packaging process.

4.4 Field Devices

The automation system interfaces with various simulated field devices that represent the equipment commonly found in an industrial bottle packaging line.

These devices include:

Sensors

- Bottle position sensors
- Filling level sensor
- Cap presence sensor

- Home position sensors
- Emergency Stop input

Actuators

- Conveyor motors
- Filling valve
- Capping actuator
- Reject mechanism
- Alarm indicator

Each field device is monitored and controlled by the PLC throughout the production cycle.

4.5 Human-Machine Interface

The Human-Machine Interface provides a graphical representation of the production process and allows operators to interact with the machine.

The HMI has been designed to provide:

- Real-time process visualization.
- Production statistics.
- Runtime monitoring.
- Machine operating status.
- Alarm indication.
- Configuration controls.
- User authentication.
- Manual fault simulation.

The interface enables operators to monitor production, adjust machine settings, and respond to abnormal operating conditions efficiently.

4.6 Production Monitoring

The system continuously records and displays key production information.

The following production indicators are available:

- Total bottles processed.

- Total rejected bottles.
- Machine runtime.
- Bottles per hour.
- Current machine status.
- Active alarm indication.

These values provide operators with a real-time overview of machine performance.

4.7 Machine Architecture

The Bottle Packaging Line follows a centralized control architecture.

The PLC acts as the central controller, receiving information from sensors, processing the required control logic, and operating field devices. The HMI communicates with the PLC to display machine status and allow operator interaction.

The architecture has been designed to provide:

- Reliable sequential operation.
 - Modular software organization.
 - Easy troubleshooting.
 - Simple future expansion.
 - Clear separation between control and visualization.
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4.8 System Boundaries

This project represents a simulated industrial automation system developed for educational and portfolio purposes.

The following features are intentionally outside the scope of the project:

- Vision inspection systems.
- Servo motion control.
- Industrial networking (EtherNet/IP, PROFINET, etc.).
- Manufacturing Execution System (MES) integration.
- SCADA integration.
- Automatic recipe management.

- Production database logging.

These features can be incorporated in future versions to more closely represent a commercial bottle packaging system.

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5. Machine interlocks

5.1 Purpose

Machine interlocks are implemented to ensure that the Bottle Packaging Line operates safely, reliably, and in the correct sequence. Interlocks prevent unintended machine operation by requiring predefined operating conditions to be satisfied before a process can begin.

The PLC continuously monitors machine status, operator commands, and process conditions before enabling equipment operation. If any required condition is not satisfied, the corresponding operation is inhibited until the condition has been corrected.

5.2 System Start Interlocks

Automatic operation shall only be permitted when all startup conditions have been satisfied.

The following conditions shall be met before the machine is allowed to start:

- The Emergency Stop is released.
- No active machine faults are present.
- All required actuators are in their normal operating position.
- The operator has pressed the Start push button.
- The machine is in Automatic Mode.

If any of these conditions are not satisfied, the system shall remain in the OFF state until corrective action has been taken.

5.3 Filling Interlocks

The filling operation shall only begin when all required filling conditions have been satisfied.

The PLC shall verify:

- A bottle is correctly positioned at the filling station.
- The conveyor has stopped.
- No filling-related fault is active.
- The machine is operating in Automatic Mode.

If any condition becomes invalid during filling, the filling operation shall be terminated and the machine shall respond according to the applicable fault condition.

5.4 Capping Interlocks

The capping operation shall only begin after successful completion of the filling operation.

Before extending the capping actuator, the PLC shall confirm:

- The bottle has reached the capping station.
- Filling has been completed.
- No active capping fault is present.
- The capping actuator is in its home position.

The bottle shall not proceed to the next station until the capping cycle has been completed successfully.

5.5 Sorting and Reject Interlocks

The sorting operation shall begin only after completion of the capping operation.

Before making a sorting decision, the PLC shall evaluate:

- Cap presence status.
- Leak bottle status.
- Reject mechanism availability.

If the bottle satisfies all quality requirements, it shall continue toward the outfeed section.

If any quality requirement is not satisfied, the bottle shall be rejected.

5.6 Alarm Interlocks

Whenever a machine fault is detected, the PLC shall apply the appropriate operating restrictions.

Depending on the fault type, the PLC may:

- Prevent machine startup.
- Stop the current process.
- Disable a specific actuator.
- Stop the complete production line.
- Require operator intervention before restarting production.

The HMI shall provide visual indication of all active alarms and allow the operator to acknowledge and reset alarms after corrective action has been completed.

5.7 Emergency Stop Philosophy

The Emergency Stop provides immediate shutdown of the machine during unsafe operating conditions.

When the Emergency Stop is activated:

- All machine outputs shall be de-energized immediately.
- Conveyor movement shall stop.
- Filling operation shall stop.
- Capping operation shall stop.
- Sorting operation shall stop.
- Production sequence shall be interrupted.
- Alarm indication shall remain active.

After the Emergency Stop has been released, the machine shall remain stopped until the operator performs the required alarm reset procedure and initiates a new Start command.

Automatic restart following an Emergency Stop is not permitted.

5.8 Operator Interlocks

Certain machine functions require operator authorization before execution.

These include:

- Modifying production settings.
- Resetting production statistics.
- Resetting runtime values.

Access to these functions shall be controlled through user authentication within the HMI.

5.9 Production Interlocks

The machine shall maintain sequential process integrity throughout the production cycle.

The PLC shall ensure that:

- Only one processing operation is active for a bottle at any given station.
- Each process completes before the next process begins.
- Bottle movement occurs only when the next station is available.
- Production statistics are updated only after completion of a bottle cycle.

These interlocks ensure stable and repeatable machine operation.

5.10 Future Interlock Expansion

The current project implements the interlocks required for the simulated bottle packaging process.

In a commercial production system, additional interlocks may include:

- Safety gate monitoring.
- Air pressure monitoring.
- Low product level detection.
- Motor overload protection.
- Conveyor jam detection.
- Reject bin full detection.
- Network communication monitoring.
- Power failure recovery.
- Servo drive status monitoring.

These features are outside the scope of the current Proof of Concept but may be incorporated in future system developments.

6. Alarm Philosophy

6.1 Purpose

The alarm system has been implemented to notify the operator whenever the machine detects an abnormal operating condition that may affect production, product quality, or machine safety.

The primary objectives of the alarm system are to:

- Inform the operator of abnormal machine conditions.
- Prevent damage to machine components.
- Prevent defective products from being accepted.
- Maintain safe machine operation.
- Assist troubleshooting and fault recovery.
- Minimize production downtime.

The PLC continuously monitors machine status and automatically generates alarms whenever predefined fault conditions are detected.

6.2 Alarm Classification

The Bottle Packaging Line uses two categories of alarms.

Process Alarms

Process alarms indicate abnormal production conditions that affect product quality but do not necessarily represent mechanical failures.

Examples include:

- Leak Bottle Fault
- Cap Missing Fault

When a process alarm occurs, the affected bottle is identified as defective and is rejected at the Combined Sorting and Reject Station.

Equipment Alarms

Equipment alarms indicate abnormal operation of machine components that prevents the production process from continuing safely.

Examples include:

- Capping Arm Retract Jam
- Capping Arm Complete Jam
- Sorting Arm Retract Jam
- Sorting Arm Complete Jam
- Emergency Stop

Equipment alarms generally require operator intervention before production can continue.

6.3 Alarm Detection

The PLC continuously evaluates machine status during every scan cycle.

Whenever a monitored condition exceeds its normal operating limits or an expected machine response is not achieved, the corresponding alarm is generated automatically.

Alarm detection is performed continuously throughout automatic machine operation to ensure immediate response to abnormal conditions.

6.4 Operator Notification

Whenever an alarm is generated, the operator shall be informed through the Human-Machine Interface.

The HMI provides:

- Active alarm indication.
- Alarm description.
- Alarm acknowledgement.
- Alarm silence function.
- Alarm reset function.

This enables the operator to quickly identify the cause of the abnormal condition and perform the required corrective action.

6.5 Machine Response

The PLC responds differently depending on the type of alarm generated.

Typical responses include:

- Rejecting the affected bottle.
- Stopping a specific machine operation.
- Stopping the complete production line.
- Preventing machine startup.
- Preventing continuation of the production sequence.
- Activating alarm indication.

The machine response is determined by the severity and type of the detected fault.

6.6 Alarm Acknowledgement

After an alarm has been generated, the operator may acknowledge the alarm through the HMI.

Alarm acknowledgement indicates that the operator has recognized the fault condition.

Acknowledging an alarm does not automatically clear the fault or restart the machine.

The underlying cause of the alarm must first be corrected.

6.7 Alarm Reset

After the fault condition has been corrected, the operator shall perform the Alarm Reset procedure.

Alarm Reset performs the following functions:

- Clears the active alarm.
- Resets internal fault indicators.
- Restores normal machine status.
- Enables the machine to be restarted.

The machine shall remain stopped until the operator issues a new Start command.

6.8 Fault Recovery Philosophy

Following correction of a machine fault, production shall resume only after:

- The fault has been corrected.
- The Emergency Stop has been released (if applicable).

- The operator acknowledges the alarm.
- The operator performs an Alarm Reset.
- The machine is confirmed to be in a safe operating condition.
- A new Start command is issued.

Automatic restart following a fault is not permitted.

6.9 Future Alarm Expansion

The current Proof of Concept includes the alarms required to demonstrate machine operation and fault handling.

Future industrial implementations may incorporate additional alarms, including:

- Conveyor Jam Detection
- Low Tank Level
- High Tank Level
- Air Pressure Low
- Motor Overload
- Sensor Failure
- Communication Failure
- Reject Bin Full
- Preventive Maintenance Reminder
- Power Failure Recovery

These additional alarms are outside the scope of the current project but may be implemented in future system enhancements.

7. Safety Philosophy

7.1 Purpose

The safety philosophy defines the fundamental principles adopted to ensure safe operation of the Bottle Packaging Line during normal production, fault conditions, and operator interaction.

The system has been developed to demonstrate industrial automation concepts while incorporating basic machine safety practices appropriate for a Proof of Concept (PoC) project.

7.2 Safety Objectives

The primary safety objectives of the system are:

- Protect personnel from unintended machine operation.
 - Prevent damage to machine components.
 - Minimize production interruptions caused by unsafe conditions.
 - Ensure controlled machine startup and shutdown.
 - Provide immediate machine stopping capability through the Emergency Stop.
 - Prevent automatic restart following a fault or emergency condition.
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7.3 Emergency Stop Function

The Emergency Stop provides the highest level of operator intervention within the system.

When activated:

- All machine outputs are immediately de-energized.
- Conveyor motion stops.
- Filling operation stops.
- Capping operation stops.
- Sorting operation stops.
- The machine enters a safe stopped condition.
- Alarm indication remains active until the fault is cleared.

The machine shall not resume operation automatically after the Emergency Stop has been released.

7.4 Controlled Startup

The machine shall only enter automatic operation after the following conditions have been satisfied:

- Emergency Stop released.

- No active machine faults.
- Alarm Reset completed.
- Operator Start command received.
- All required machine conditions verified.

This ensures that production begins only under safe operating conditions.

7.5 Controlled Shutdown

The operator may stop production using the Stop command.

During a normal shutdown:

- Active processing operations are completed where appropriate.
 - Machine outputs are de-energized in a controlled manner.
 - The production sequence is terminated.
 - The machine returns to the OFF state.
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7.6 Proof of Concept Safety Statement

This project has been developed as an educational Proof of Concept and does not include the complete safety systems required for commercial industrial machinery.

Industrial production systems typically incorporate additional safety measures such as:

- Safety PLCs
- Safety relays
- Dual-channel Emergency Stop circuits
- Safety-rated sensors
- Safety gate monitoring
- Lockout/Tagout procedures
- Functional safety standards (ISO 13849, IEC 62061, IEC 61508)

These features are beyond the scope of the current implementation but should be considered essential for deployment in a real industrial environment.

8. Performance Requirements

8.1 General Performance

The Bottle Packaging Line shall provide stable and repeatable operation during automatic production. The control system shall continuously monitor machine status, execute the production sequence, and respond to operator commands and fault conditions in a predictable manner.

The system shall maintain sequential processing of bottles while ensuring that each station completes its assigned operation before the bottle proceeds to the next stage.

8.2 Production Performance

The system shall provide continuous monitoring of production performance through the HMI.

The following production information shall be available to the operator:

- Total bottles processed
- Total rejected bottles
- Machine runtime
- Bottles per hour
- Current operating status
- Active alarm indication

Production statistics shall be updated automatically during machine operation.

8.3 Reliability

The machine shall operate using a deterministic PLC scan cycle to ensure consistent process execution.

The modular software structure shall simplify troubleshooting, maintenance, and future program expansion while maintaining reliable machine operation.

8.4 Availability

The system shall be designed to minimize unnecessary production interruptions by:

- Providing clear alarm messages.
- Allowing rapid fault identification.

- Supporting controlled fault recovery.
 - Enabling quick machine restart after corrective action.
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9. Design Assumptions

The design of this project is based on the following assumptions:

- Bottles arrive individually at each processing station.
- Sensors operate correctly unless a fault is intentionally simulated.
- Pneumatic actuators complete their movement within the expected operating time.
- The filling process reaches the configured preset level without interruption.
- The communication between the PLC and HMI remains available during normal operation.
- The operator follows the correct machine startup and shutdown procedures.
- The machine operates in a simulated environment without external disturbances.

These assumptions provide the basis for the functional behaviour described in this document.

10. Project Limitations

This project has been developed as a Proof of Concept to demonstrate industrial automation principles and does not represent a complete commercial bottle packaging system.

The following features are outside the scope of the current implementation:

- Vision inspection systems.
- Barcode or QR code tracking.
- Servo-controlled positioning systems.
- Variable Speed Drive (VFD) control.
- Automatic recipe management.
- Industrial communication protocols such as EtherNet/IP, PROFINET, or OPC UA.
- Manufacturing Execution System (MES) integration.
- SCADA integration.
- SQL database production logging.

- Safety PLC implementation.
- Redundant control architecture.
- Predictive maintenance functionality.
- Energy monitoring and reporting.

These features may be incorporated into future versions of the project.

11. Future Expansion

The modular architecture adopted within this project allows additional functionality to be integrated with minimal modification to the existing control structure.

Potential future enhancements include:

- Vision-based quality inspection.
- Automatic bottle size selection.
- Multiple filling recipes.
- Servo motion control.
- Conveyor speed control using Variable Speed Drives.
- Industrial Ethernet communication.
- OPC UA connectivity.
- MQTT-based IIoT communication.
- Cloud-based production monitoring.
- SQL database integration.
- Overall Equipment Effectiveness (OEE) monitoring.
- Energy consumption analysis.
- Predictive maintenance.
- Multi-user security management.
- Remote monitoring through web dashboards.
- Integration with SCADA and MES platforms.

These enhancements would further align the project with modern Industry 4.0 manufacturing practices.

12. Glossary

Term	Description
PLC	Programmable Logic Controller
HMI	Human-Machine Interface
PoC	Proof of Concept
KPI	Key Performance Indicator
Ladder Logic	Graphical PLC programming language
Reject Station	Station responsible for removing defective bottles
Runtime	Total machine operating time
Cycle	One complete bottle processing sequence
Fault	Abnormal operating condition requiring operator attention

13. References

The following software, hardware, and technical resources were used during the development of this project:

Hardware

- Allen-Bradley MicroLogix 1100 PLC

Software

- RSLogix 500
- RSLogix Emulate 500
- RSLinx Classic
- Wonderware InduSoft Web Studio (AVEVA Edge)

Standards and Engineering Practices

- Modular PLC Programming Principles
- Sequential Machine Control
- Industrial Human-Machine Interface Design Practices

- General Industrial Automation Engineering Concepts
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14. Conclusion

The Functional Design Specification defines the intended operation and functional behaviour of the Industrial Bottle Packaging Line Automation project.

The document establishes the system objectives, operating philosophy, functional requirements, machine interlocks, alarm philosophy, safety philosophy, and overall design considerations that form the foundation of the automation system.

The project demonstrates the implementation of modular PLC programming, Human-Machine Interface development, process automation, and fault management within a simulated bottle packaging environment.

Although developed as a Proof of Concept, the project reflects commonly adopted industrial automation principles and provides a scalable foundation for future enhancements and integration with advanced manufacturing technologies.